Zero-Shot Transfer in Multi-Omics Breast Cancer Detection with GraphAny

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Abstract

Multi-omics datasets such as TCGA-BRCA, METABRIC, and CPTAC provide rich information for cancer subtype prediction but differ strongly in feature spaces and preprocessing, causing most graph neural networks (GNNs) to require retraining and to often degrade under distribution shift. We investigate whether GraphAny, a fully inductive GNN combining analytic LinearGNN solvers with an attention-fusion mechanism, can better handle cross-cohort variation. We will construct patient similarity graphs and evaluate GraphAny in two settings: in-domain prediction on TCGA-BRCA and zero-shot transfer to METABRIC and CPTAC, and compare GraphAny to standard GNN and transformer baselines and assess robustness under platform and modality shift.

1 Introduction

Breast cancer subtyping is essential for precision oncology, as treatment decisions depend on reliable molecular classification. We use PAM50 as the target label in our study [11]. Recent works in breast cancer histopathology use foundation model features with graph autoencoders to achieve state-of-the-art retrieval performance [13], but these models remain modality-specific and are not evaluated under cross-dataset distribution shift or multi-omics settings [2]. Large cohorts such as TCGA-BRCA and METABRIC, and a modality shift such as CPTAC provide multi-omics profiles (RNA-seq, methylation, CNV, miRNA, proteomics), offering rich information for predictive modeling. However, these datasets differ in platforms, preprocessing, and subtype definitions, making cross-cohort prediction difficult. A model that performs consistently across heterogeneous datasets would reduce repeated retraining and support more robust clinical translation.

Most multi-omics GNNs treat each cohort as a separate task. Models such as GCN, GATv2, MOGAT, and DeepMAPS must be retrained for each dataset because they assume identical feature spaces and distributions. These methods struggle when modalities or feature scales differ and are typically evaluated only in-domain, without testing zero-shot transfer [17, 18]. This limits practical use: real deployment requires robustness across patient populations and sequencing platforms. Existing reviews highlight that cross-dataset generalization remains largely untested [6, 14], leaving an open gap for models that transfer across cohorts without retraining.

Existing GNNs tightly couple node features and graph structure, making them sensitive to shifts in feature dimension, modality composition, and subtype definitions. Their message passing does not account for entropy or scale differences across datasets [7, 14]. Even attention-based or transformer models for omics rely on fixed feature encodings. To our knowledge, no study has tested whether a single GNN can be trained on one cohort and directly applied to another without retraining [14].

We investigate GraphAny, a fully-inductive GNN that combines analytic LinearGNN solvers with an attention-fusion mechanism designed for heterogeneous feature spaces. GraphAny fuses solver outputs using entropy-normalized distance features, enabling generalization to unseen datasets. We evaluate applying this model to multi-omics breast cancer subtype prediction by representing each patient as a node in a similarity graph, an approach shown to improve predictive performance in prior studies [3, 5, 8, 19].

Below, we summarize our contributions.

- Evaluate GraphAny for multi-omics subtype prediction on TCGA-BRCA and compare it to strong GNN baselines.
- Test zero-shot transfer from TCGA-BRCA to METABRIC and CPTAC against baselines requiring retraining.
- Provide a controlled experimental setup for cross-dataset evaluation with standardized preprocessing and metrics.

2 Related Work

Research on multi-omics cancer prediction has increasingly explored graph neural networks (GNNs), but existing work differs in scope, datasets, and evaluation practices. In parallel, graph-based models have also been applied to breast cancer histopathology using features from medical foundation models, further highlighting the potential of combining graphs with powerful pre-trained representations in oncology. We summarize four main areas relevant to this study.

2.1 GNNs for Multi-Omics Subtype Prediction

GNNs such as omicsGAT [1], and MOGAT [15] use patient similarity graphs to integrate multi-omics data and report strong classification performance on single cohorts. Similar approaches construct sample similarity networks [3]. However, these methods are typically evaluated only in-domain and depend on cohort-specific preprocessing pipelines [1, 15]. To our knowledge, cross-dataset transfer is still rarely evaluated in bioinformatics modeling, leaving generalization under distribution shift unclear [14].

2.2 Graph Transformers in Biology

Graph transformers such as DeepMAPS [9] and GraphGPS [12] extend message passing with global attention, and have shown strong performance across diverse graph benchmarks. However, these models typically assume fixed feature spaces and are commonly evaluated in transductive settings, which limits their suitability for cross-cohort multi-omics transfer.

2.3 Foundation Models in Breast Cancer

Recent work on breast histopathology shows that combining medical foundation model features with graph-based models can substantially improve downstream performance. Saeidi et al. propose an attention-based adversarially regularized variational graph autoencoder for breast histology image retrieval

and demonstrate that using features from pathology foundation models such as UNI yields significantly higher mAP and mMV scores on BreakHis and BACH compared to conventional CNN features [13]. Reviews of graph neural networks in histopathology similarly report that pathology foundation models provide stronger slide- and patch-level representations than traditional CNNs, but these advances remain confined to imaging and do not address cross-cohort multi-omics subtype prediction at the patient-graph level [2].

2.4 GraphAny and Inductive OOD Robustness

Out-of-Distribution(OOD) generalization is increasingly studied in bioinformatics [14], but rarely for graphs. GraphAny [19] is a fully-inductive GNN designed to handle arbitrary graphs with differing feature spaces through analytic solvers and attention fusion. Multiple LinearGNN variants are combined via an inductive attention fusion module that uses distance-based agreement signals between solver outputs. This design makes the fusion step independent of the specific feature dimension of a target graph and supports transfer to unseen graphs [19]. While promising, it has not been evaluated on biomedical or multi-omics datasets. Current OOD studies also lack standardized evaluation protocols, making it difficult to assess true generalization performance [7].

2.5 Summary and Reflection

Existing GNN and transformer models in breast cancer classification perform well within single datasets but do not address cross-cohort variability, use inconsistent preprocessing, and largely ignore zero-shot transfer. Inductive models like GraphAny offer a potential path toward robustness across heterogeneous multi-omics data. Our work fills this gap by evaluating GraphAny on patient similarity graphs, training on TCGA-BRCA and then testing zero-shot generalization across METABRIC, and CPTAC under a controlled and reproducible setup.

3 Experimental Apparatus

3.1 Datasets

We evaluate on three public breast cancer cohorts that differ in measurement technology and molecular modality. TCGA-BRCA is used for in-domain training and evaluation, while METABRIC and CPTAC are used exclusively for zero-shot transfer. Dataset access details and file lists are provided in Appendix A.1.

TCGA BRCA. TCGA-BRCA includes RNA-seq gene expression, DNA methylation, and gene-level copy number variation [16]. Patients are identified by TCGA barcodes, and modalities are aligned by barcode matching.

METABRIC. The METABRIC breast cancer provides gene expression profiles measured by microarray together with clinical annotations [4]. The use of microarray-based expression introduces a platform shift relative to TCGA RNA-seq.

CPTAC BRCA. CPTAC-BRCA consists of protein abundance measurements derived from mass spectrometry, along with clinical metadata [10]. This dataset introduces both a modality shift (proteomics vs. transcriptomics) and a platform shift relative to TCGA and METABRIC.

Target label. We predict breast cancer intrinsic subtype using the PAM50 classification [11], which defines five molecular subtypes. PAM50 is consistently available across cohorts, enabling controlled cross-cohort evaluation.

Table 1: Cohort overview after preprocessing. N is the number of retained patients; Cls is the number of PAM50 classes.

Cohort	N	Modalities	level	Cls
TCGA BRCA	≈ 700	RNA+Meth+CNV	Gene	5
METABRIC	≈ 1900	Expr	Gene	5
CPTAC BRCA	≈ 100	Prot	Protein	5

Modalities: RNA = RNA-seq, Meth = DNA methylation, CNV = copy number variation, Expr = microarray expression, Prot = protein abundance.

Descriptive statistics. Table 1 reports approximate cohort sizes and will be finalized after preprocessing and modality intersection.

3.2 Baselines.

We compare GraphAny against representative GNN baselines used in prior multi-omics cancer studies. These include standard message-passing models (*GATv2 [17]*), multi-omics patient-graph models (*omicsGAT* [1], *MOGAT* [15]), and a transformer-style baseline (*GraphGPS [12]*) reflecting recent trends in biological graph learning. All baselines are evaluated under the same splits and metrics as GraphAny for in-domain experiments. For cross-cohort evaluation, baselines are retrained on the target cohort and then evaluated.

3.3 Preprocessing

We align samples by patient identifiers, retain only labeled patients, and transform each omics table into a standardized patient-by-feature matrix. Features are filtered for missingness and low variability, missing values are imputed (median), and all scaling parameters are fit on the TCGA training split and reused unchanged for zero-shot cohorts to avoid target-cohort leakage. Label information is never used during feature preprocessing or graph construction, and only the training split labels are used for model fitting.

3.4 Procedure

Our experimental procedure is designed to evaluate model performance under both in-domain and cross-cohort settings.

Graph construction. For each cohort, we construct a patient similarity graph where nodes represent patients and edges connect similar patients based on cosine similarity of their concatenated modality features. Patient similarity graphs have been used for cancer subtype prediction in prior work [3]. We build a k-nearest neighbor graph with fixed k, symmetrize the edges, and scale the weights to $[0, 1]$. Additional justification for cosine similarity and kNN graph construction is given in Appendix A.3.

Task formulation. Although diagnosis is often framed as graph classification, our setting is naturally node classification because the cohort forms a single patient graph and the model predicts a label for each patient node. This formulation lets the model exploit relationships between patients and is well suited for small cohort sizes and high dimensional omics features. Node features are patient-level omics vectors (one row per patient), which are used directly as $\mathbf{X}$ for GraphAny and all baselines.

Hyperparameter optimization. we search over learning rates {3e-4, 1e-3, 3e-3}, weight decay {1e-5, 1e-4, 5e-4}, hidden sizes {64, 128, 256}, layers {2, 3}, and dropout {0.2, 0.5}. Models are trained

with Adam for up to 200 epochs using early stopping (patience 20) based on validation macro F1. Selected hyperparameters are reused for each model during retraining on target cohorts. The full hyperparameter search space is reported in Appendix A.2.

In-domain evaluation (RQ1). We evaluate all models on TCGA-BRCA using an 80/10/10 stratified split across training, validation, and test sets. Models are trained on the training graph, tuned on the validation set, and evaluated on the test set. We repeat the process with multiple random seeds and report the mean and standard deviation.

Zero-shot transfer (RQ2). We evaluate GraphAny in a zero-shot setting by applying the model trained on TCGA-BRCA directly to METABRIC and CPTAC graphs without any retraining or tuning. Baselines are retrained separately on each target cohort using the same protocol as in-domain evaluation.

Evaluation metrics. We report macro F1 as the primary metric due to class imbalance, along with accuracy. ROC AUC (one-vs-rest) is reported where applicable. For TCGA-BRCA, results are computed on the test set; for METABRIC and CPTAC, we evaluate on the full labeled graph.

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A Supplementary Materials

A.1 Dataset access and file list

We use three public cohorts: TCGA-BRCA, METABRIC, and CPTAC-BRCA. Below we list the exact sources and the files used to construct patient-level feature matrices and labels.

TCGA-BRCA (UCSC Xena). We downloaded the following tables from the UCSC Xena browser:

- RNA-seq expression: TCGA.BRCA.sampleMap_HiSeqV2.gz
- DNA methylation: TCGA.BRCA.sampleMap_HumanMethylation450.gz
- Gene-level copy number: TCGA.BRCA.sampleMap_Gistic2_CopyNumber_Gistic2_all_data_by_genes.gz
- Clinical annotations: TCGA.BRCA.sampleMap_BRCA_clinicalMatrix

Samples are matched by TCGA barcodes.

METABRIC (cBioPortal). We downloaded the METABRIC study bundle from cBioPortal:

- Study archive: `brca_metabric.tar.gz`

We use the expression matrix as features and the clinical tables for labels and metadata.

CPTAC-BRCA (Proteomic Data Commons). We used the TCGA-linked CPTAC breast cancer study (PDC000173) from the Proteomic Data Commons:

- Protein abundance (iTRAQ): `TCGA_Breast_BI_Proteome.itraq.tsv`
- Clinical bundle: `PDC_study_clinical_data_01142026_113832.zip`
- File manifest: `PDC_file_manifest_01142026_114218.tsv`

A.2 Hyperparameter search space

We would tune all models on TCGA-BRCA using the validation split of the 80/10/10 stratified train/validation/test partition. For each model family, we select the configuration that maximizes validation macro F1 and then report test results using the selected configuration.

Optimization and training. We would train with Adam for up to 200 epochs and apply early stopping with patience 20 based on validation macro F1. For all runs we would repeat experiments with multiple random seeds and report mean and standard deviation.

Search space. We would use the following shared grid for all models unless a model requires an additional method-specific parameter:

- Learning rate: {3e-4, 1e-3, 3e-3}
- Weight decay: {1e-5, 1e-4, 5e-4}
- Hidden size: {64, 128, 256}
- Number of layers: {2, 3}
- Dropout: {0.2, 0.5}

Transfer protocol. For GraphAny, the selected TCGA-BRCA hyperparameters would be reused unchanged for zero-shot evaluation on METABRIC and CPTAC. Baselines would also be retrained on each target cohort with the same search space and selection rule.

A.3 Graph construction choices

Each cohort would be represented as a patient similarity graph. Nodes are patients and edges connect patients with similar molecular profiles in that cohort.

Why a k-nearest neighbor graph. A kNN graph yields a sparse and controlled graph structure, avoids overly dense similarity graphs, and ensures each patient has a comparable number of neighbors. This is important for stable message passing and for comparability across cohorts with different sample sizes.

Why cosine similarity. Cosine similarity compares patients by the direction of their feature vectors, which is often more robust than magnitude-based distances in high-dimensional omics data. It reduces sensitivity to global scaling differences across platforms, which is especially relevant when transferring from RNA-seq (TCGA) to microarray (METABRIC) and to proteomics (CPTAC).

Edge symmetrization and weights. We build a directed kNN graph and then symmetrize it to obtain an undirected graph: an edge between patients i and j is kept if either i is among the k nearest neighbors of j or vice versa. Edge weights are the cosine similarity scores scaled to $[0, 1]$ within each cohort, so that message passing uses non-negative and comparable weight ranges.